RESQ

Offline-First Disaster Communication Network

SE-AAT — Model Diagrams Report

Submitted as part of Software Engineering AAT

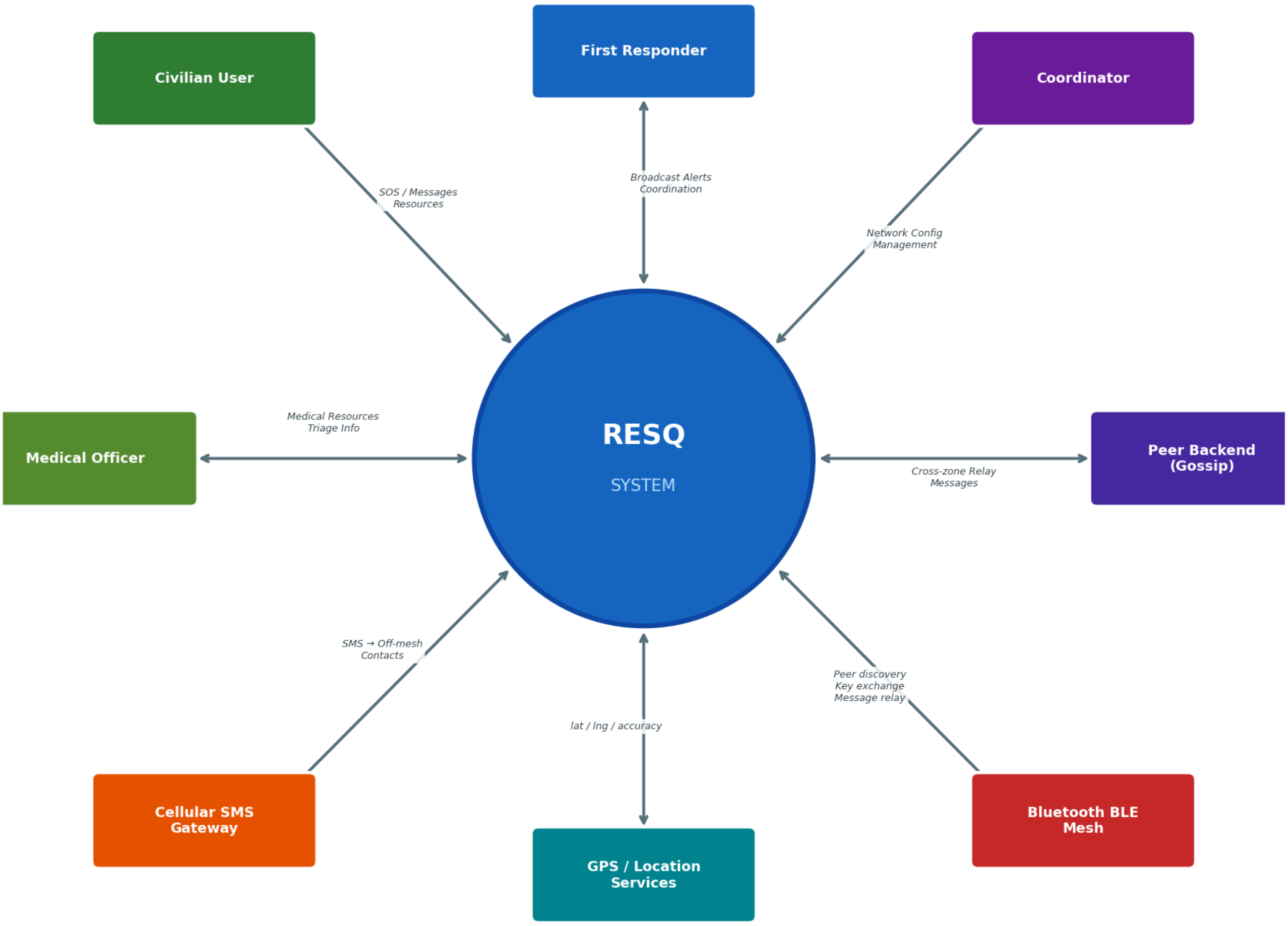
Project: Bluetooth-mesh-based offline-first communication app for disaster scenarios. Supports End-to-End Encrypted (E2EE) messaging, SOS broadcasts, resource sharing, store-and-forward queuing, and SMS to off-grid contacts. User roles: Civilian, First Responder, Medical Officer, Coordinator.

1. Context Diagram

Shows the RESQ system boundary with 8 external entities: four user roles, BLE mesh network, Cellular SMS gateway, GPS/Location services, and Peer Backend gossip node. Bidirectional arrows show the data flows into and out of the system.

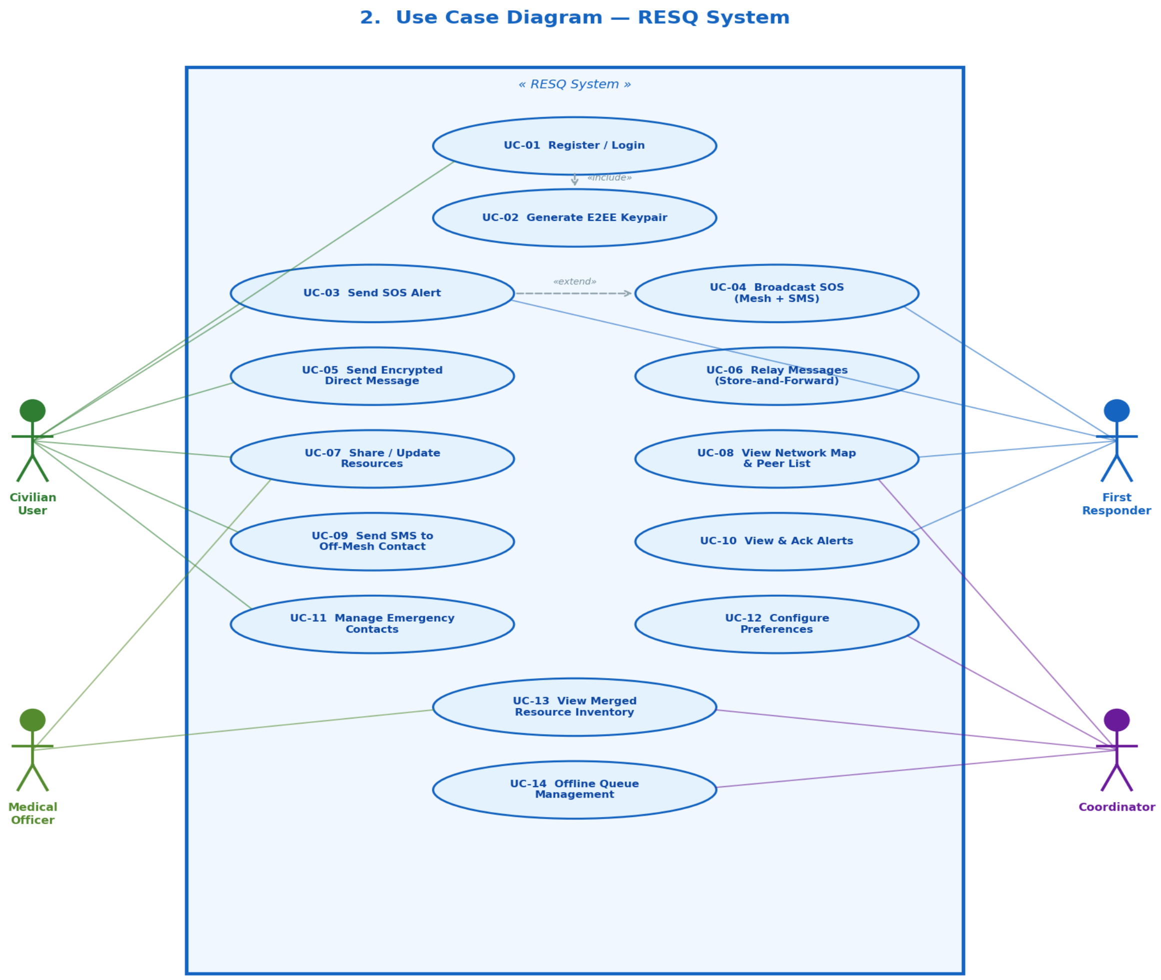
---

1. Context Diagram — RESQ Disaster Communication System



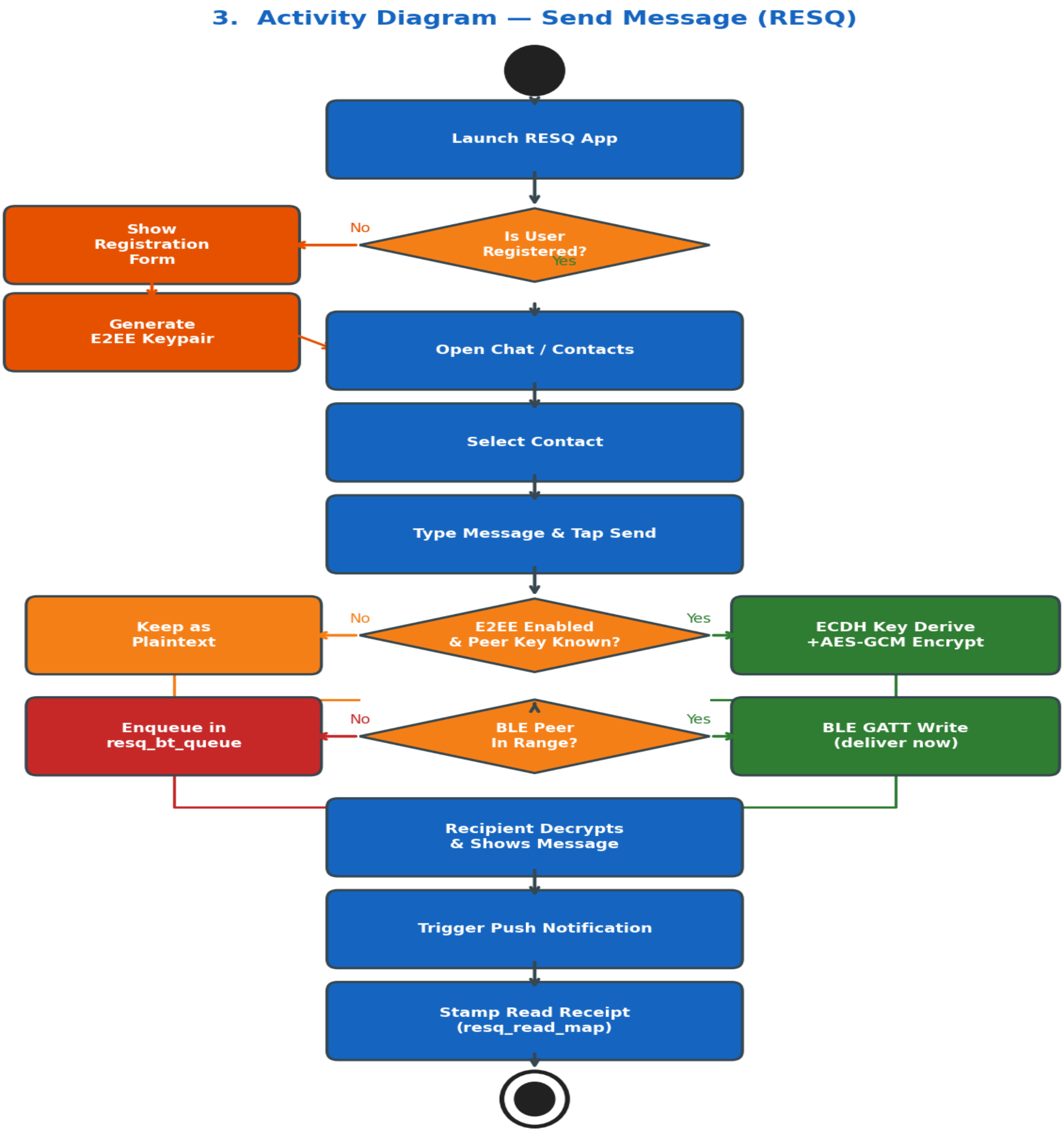
2. Use-Case Diagram

4 actors (Civilian, First Responder, Medical Officer, Coordinator) linked to 14 labelled use cases. Includes «include» relationship (Register → Generate E2EE Keypair) and «extend» (Send SOS → Broadcast via Mesh + SMS).



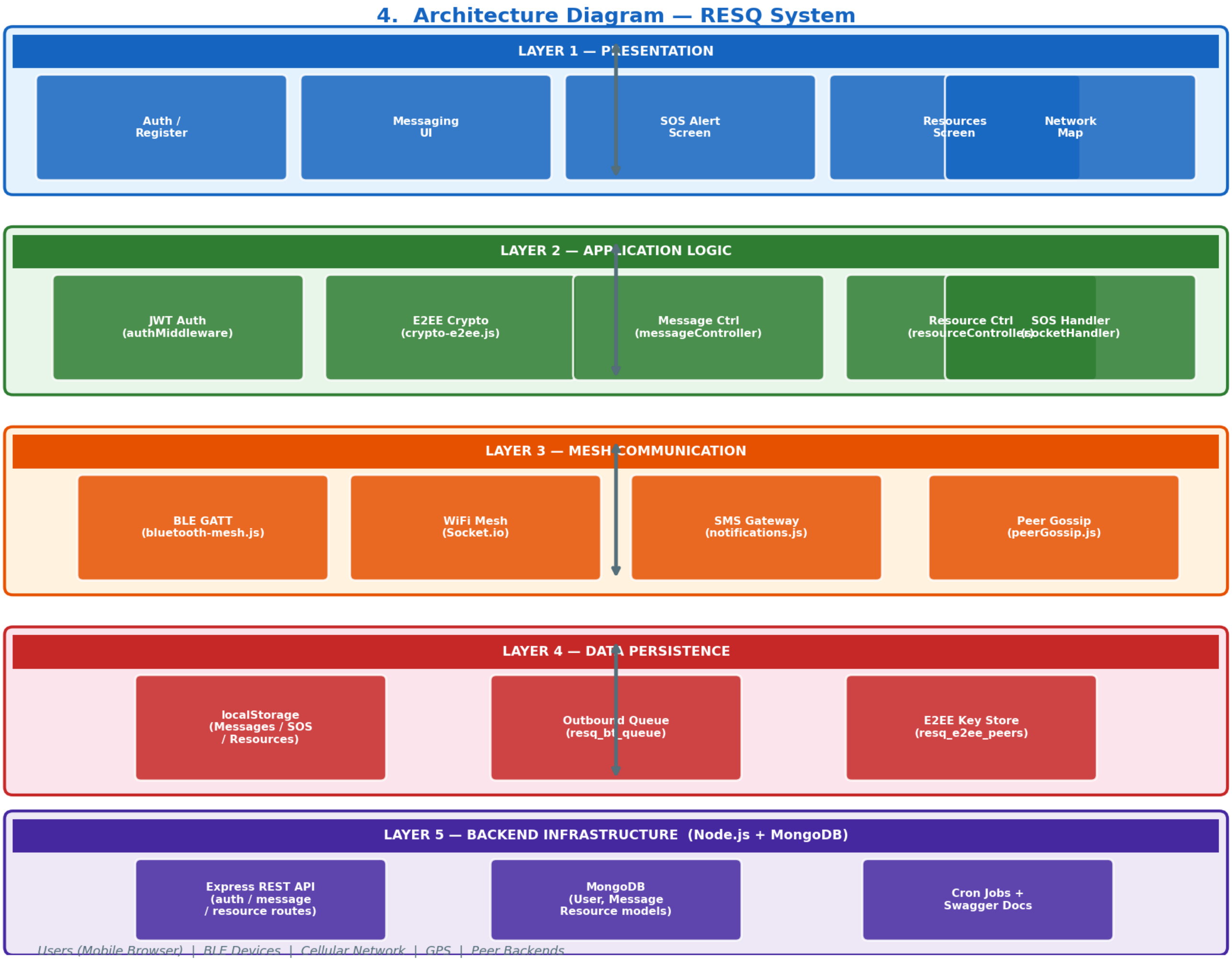
3. Activity Diagram — Send Message Flow

Complete activity flow: app launch → registration check → keypair generation → contact selection → E2EE decision → BLE delivery or queue fallback → recipient decryption → push notification → read receipt.



4. Architecture Diagram

5-layer architecture: Presentation (HTML/JS), Application Logic (Auth, Crypto, Controllers), Mesh Communication (BLE GATT, Socket.io, SMS, Gossip), Data Persistence (localStorage, Queues), and Backend Infrastructure (Node.js, Express, MongoDB).

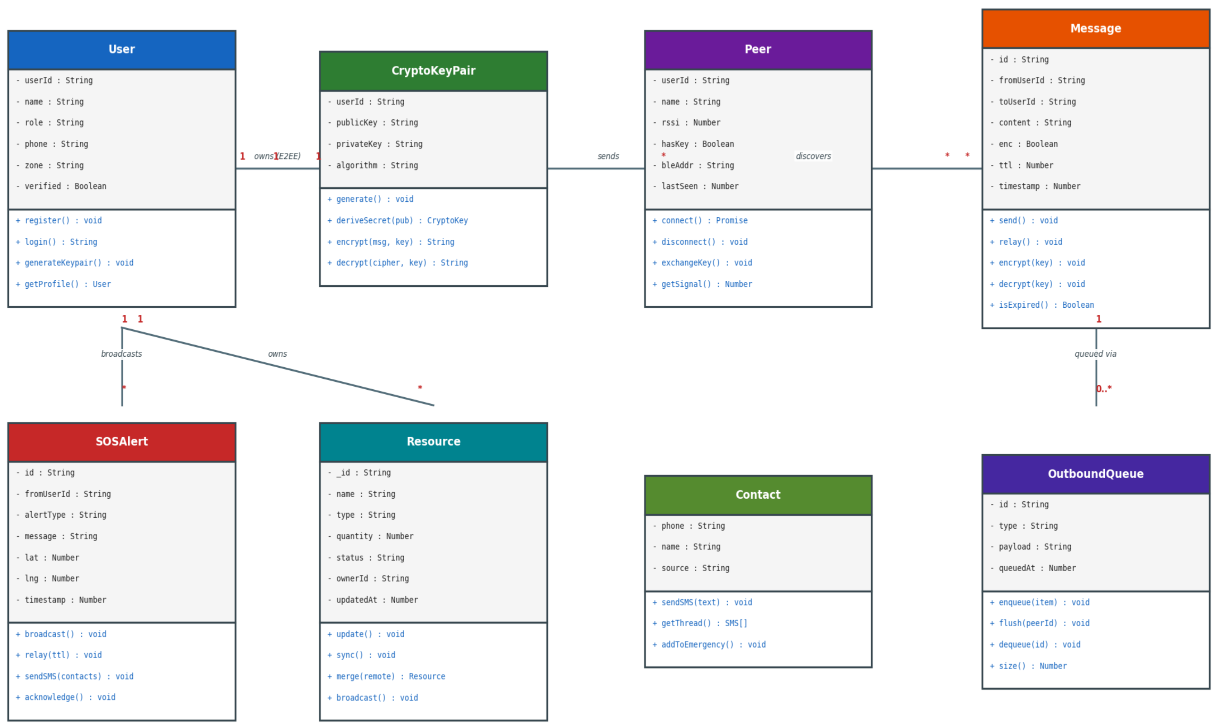


5. Class Diagram

8 UML classes with typed attributes and methods: User, CryptoKeyPair, Peer, Message, SOSAlert, Resource, Contact, OutboundQueue. Association lines show 1:1, 1:N, and M:N multiplicity.

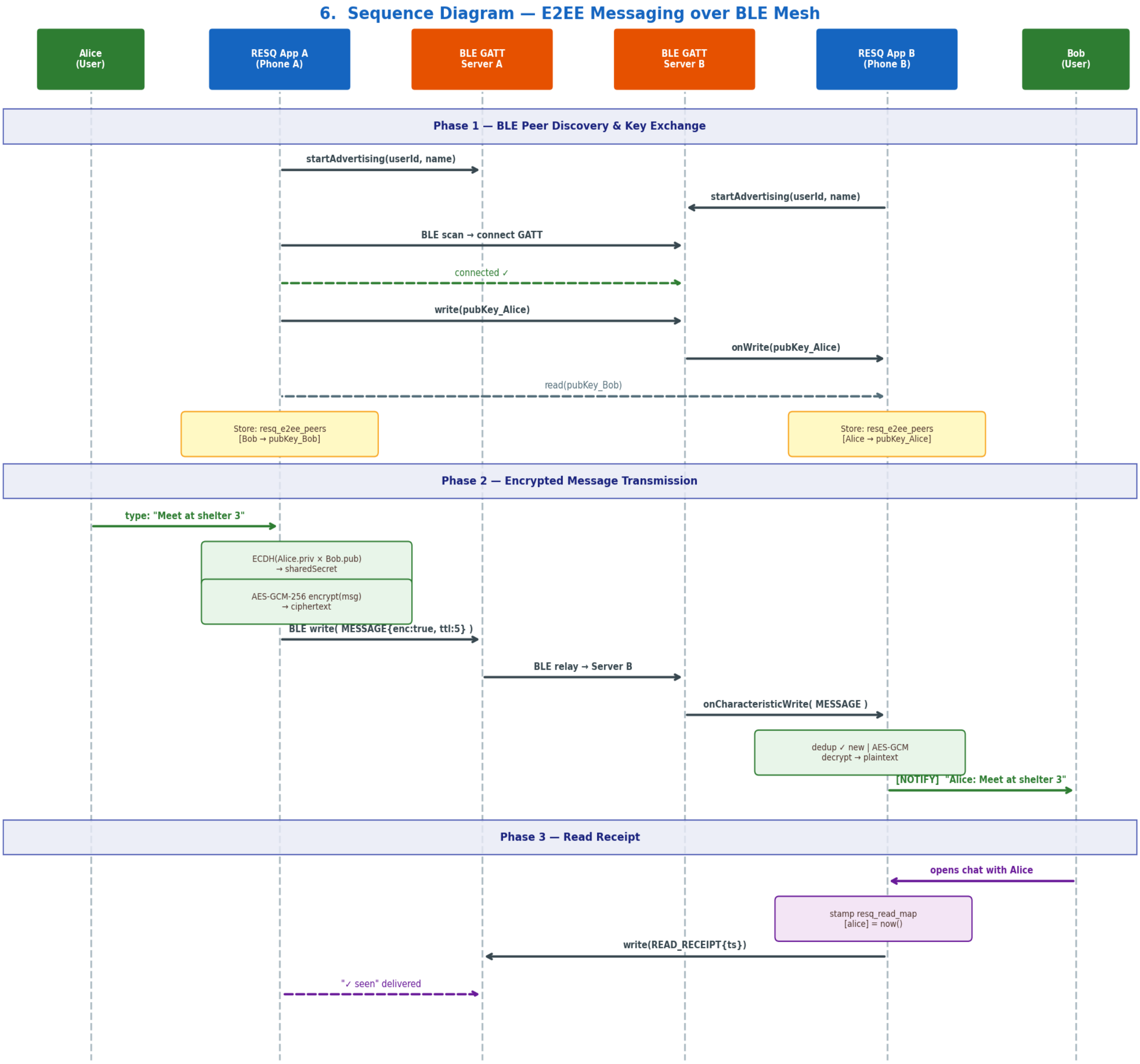
---

5. Class Diagram — RESQ System



6. Sequence Diagram — E2EE Messaging over BLE

Three phases: (1) BLE peer discovery + ECDH public-key exchange, (2) ECDH secret derivation + AES-GCM-256 encrypt → BLE write → decrypt → push notification, (3) read-receipt stamping and delivery confirmation.



7. Project Cost Estimation

Two techniques: (A) COCOMO II Basic — 15 KLOC organic, E = 41.2 PM, ~12 months, INR 24.72 lakhs; (B) Function Point (Albrecht) — 183 AFP, 12.2 PM, INR 7.32 lakhs (dev core). Phase cost bar chart and effort pie chart included.

---

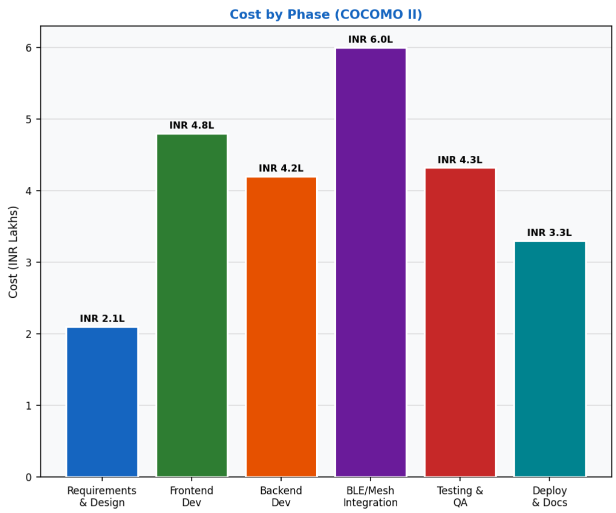
7. Project Cost Estimation — COCOMO II & Function Point Analysis

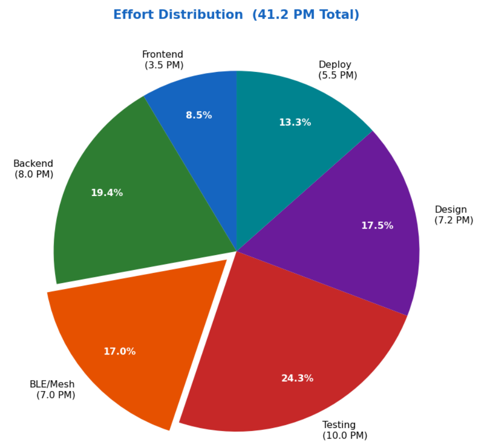
COCOMO II — Basic Model

Parameter	Value / Calculation
Project Type	Organic (small team, familiar domain)
Estimated Size	15 KLOC (JS frontend + Node.js backend)
Formula	E = 2.4 × (KLOC)^1.05
Effort E	2.4 × (15)^1.05 = 41.2 Person-Months
Dev Time D	2.5 × (E)^0.38 = 11.8 Months
Avg Team Size	N = E / D ≈ 3-4 Persons
Avg Salary (India)	INR 60,000 / person / month
Total Cost (COCOMO)	INR 41.2 × 60,000 = INR 24,72,000

Function Point (Albrecht) Estimation

Function Type	Count	Wt	FP
External Inputs (EI)	12	4	48
External Outputs (EO)	8	5	40
External Inquiries (EQ)	6	4	24
Internal Logical Files (ILF)	5	10	50
External Interface Files (EIF)	3	7	21
Unadjusted FP (UFP)			183
Complexity Adj. Factor		1.0	
Adjusted FP (AFP)			183
Productivity		15 FP/PM	
Effort (AFP / Prod.)			12.2 PM
Cost @ INR 60 K/PM			INR 7,32,000





8. Project Effort Estimation — Gantt Chart

16 tasks across 6 phases over 24 weeks (12 months): Planning (Wk 0–6), Frontend + BLE Dev (Wk 6–16), Backend Dev (Wk 6–16), Integration (Wk 13–17), Testing & QA (Wk 16–23), Deployment & Docs (Wk 18–24). Total: 41.2 person-months, 4-person team.